

1 Operating System

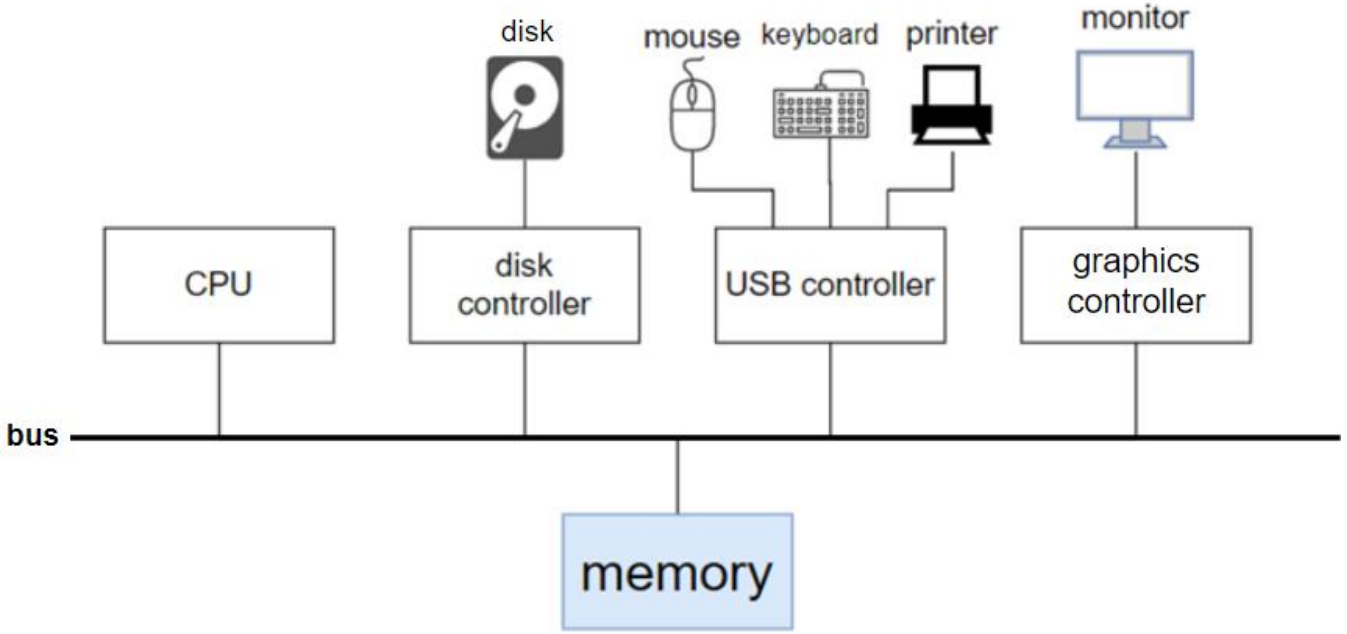
Introduction

- Ignored for a while
- Suddenly headlines
- OSs are everywhere
- Lots of examples
- Avoid religious wars
- OS critical to application success

Computer-System Organization

Computer-system organization

- One or more CPUs, device controllers connect through common bus providing access to shared memory
- Concurrent execution of CPUs and devices competing for memory cycles



Terms & Concepts

- **Device - piece of hardware connected to the main computer hardware.**

- **Device controller - electronic interface which controls devices connected to the computer.**

- **Bus - conductors connecting many hardware components in a computer.**

Computer-System Operation

- I/O devices and the CPU can execute concurrently
- Each device controller is in charge of a particular device type
- Each device controller has a local buffer
- CPU moves data from/to main memory to/from local buffers
- I/O is from the device to local buffer of controller
- Device controller informs CPU that it has finished its operation by causing an interrupt

Common Functions of Interrupts

- Interrupt transfers control to the interrupt service routine generally, through the interrupt vector, which contains the addresses of all the service routines
- Interrupt architecture must save the address of the interrupted instruction
- A trap or exception is a software-generated interrupt caused either by an error or a user request
- An operating system is interrupt driven

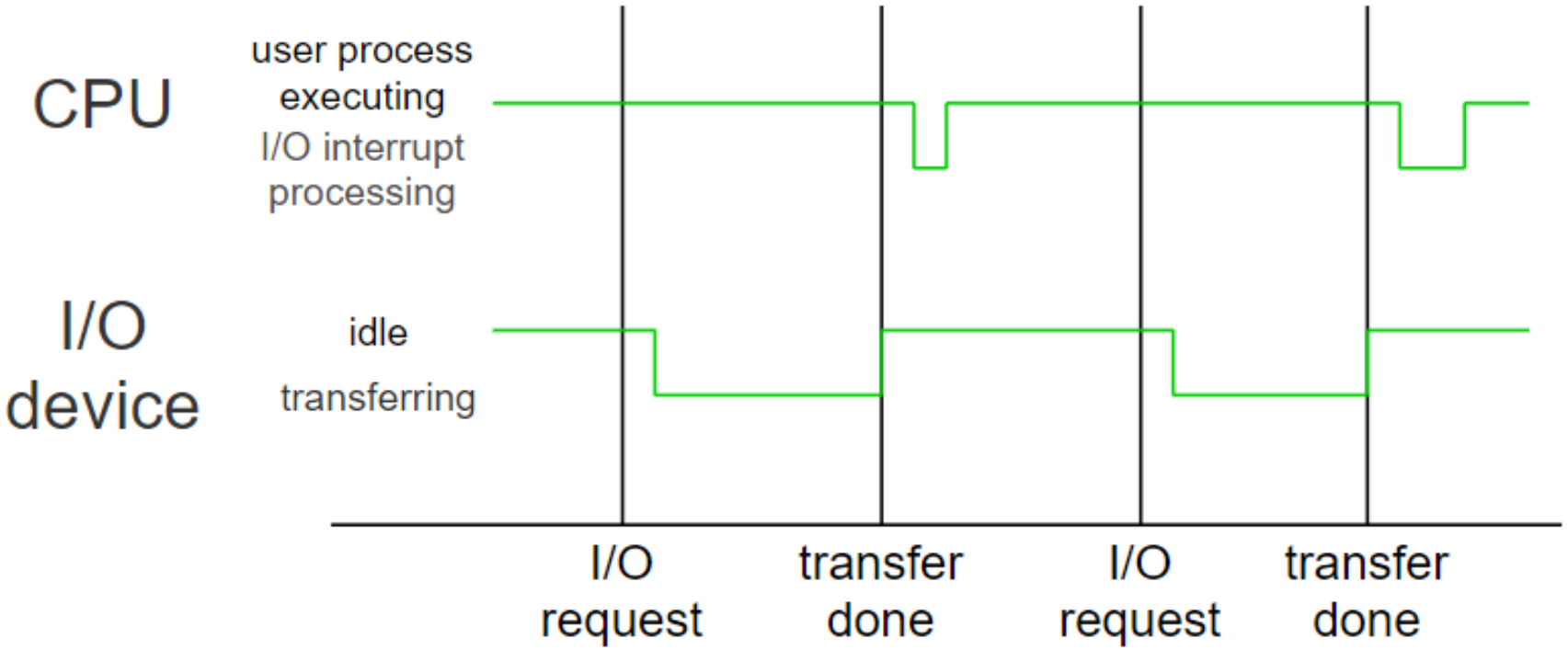
Interrupt Handling

- **The operating system preserves the state of the CPU by storing registers and the program counter**

- **Determines which type of interrupt has occurred:**
  - **polling**
  - **vectored** interrupt system

- **Separate segments of code determine what action should be taken for each type of interrupt**

Interrupt Timeline



I/O Structure

- **After I/O starts, control returns to user program only upon I/O completion**
  - Wait instruction idles the CPU until the next interrupt
  - Wait loop (contention for memory access)
- **After I/O starts, control returns to user program without waiting for I/O completion**
- **At most one I/O request is outstanding at a time, no simultaneous I/O processing**

Direct Memory Access Structure

- **Used for high-speed I/O devices able to transmit information at close to memory speeds**
- **Device controller transfers blocks of data from buffer storage directly to main memory without CPU intervention**
- **Only one interrupt is generated per block, rather than the one interrupt per byte**

Storage Structure

- **Main memory – only large storage media that the CPU can access directly**
  - **Random access**
  - Typically **volatile**

- **Secondary storage – extension of main memory that provides large nonvolatile storage capacity**

Storage Structure

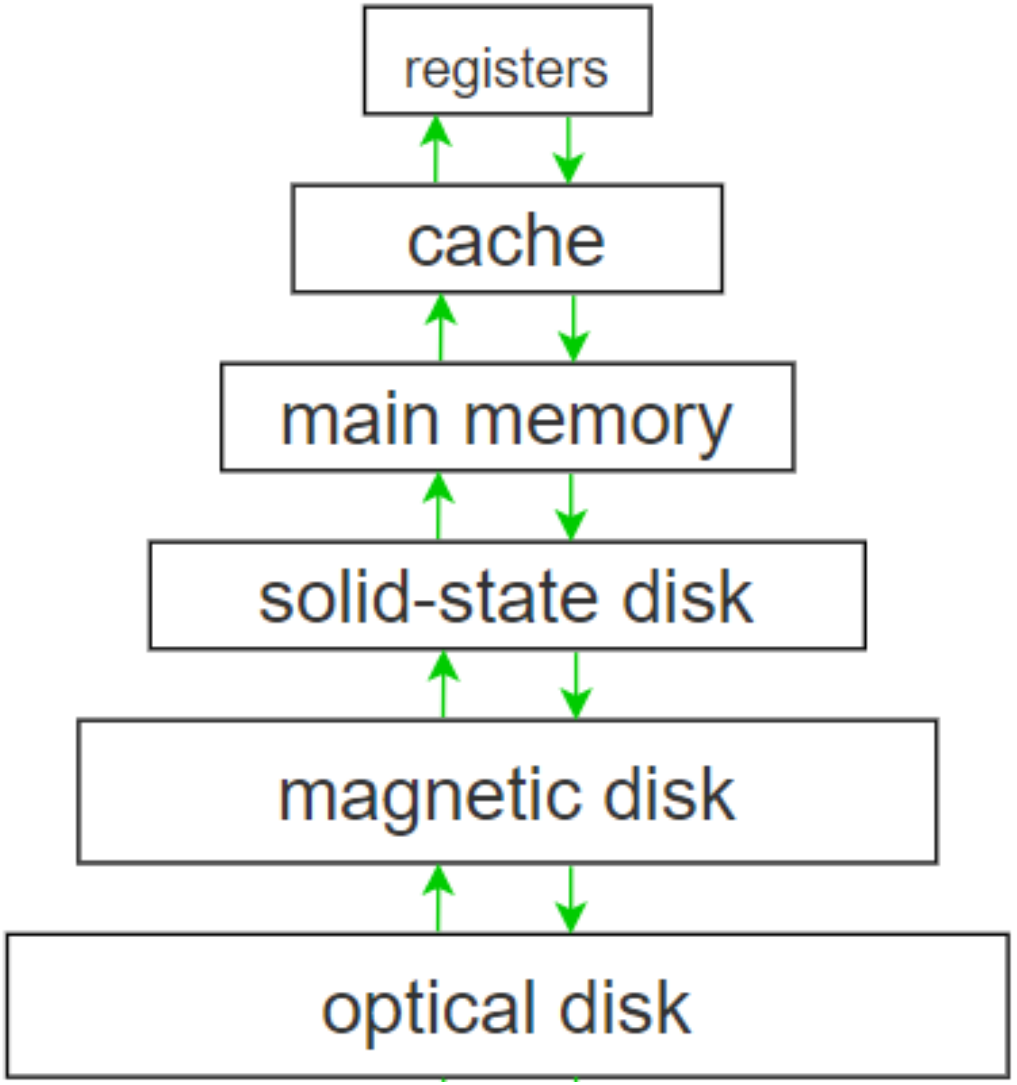
- **Magnetic disks – rigid metal or glass platters covered with magnetic recording material**
  - Disk surface is logically divided into **tracks**, which are subdivided into **sectors**
  - The **disk controller** determines the logical interaction between the device and the computer
- **Solid-state disks – faster than magnetic disks, nonvolatile**
  - Various technologies
  - Becoming more popular

Storage Hierarchy

- **Storage systems organized in hierarchy**
  - Speed
  - Cost
  - Volatility

- **Caching – copying information into faster storage system; main memory can be viewed as a cache for secondary storage**
- **Device Driver for each device controller to manage I/O**
  - Provides uniform interface between controller and kernel

Storage-Device Hierarchy



Caching

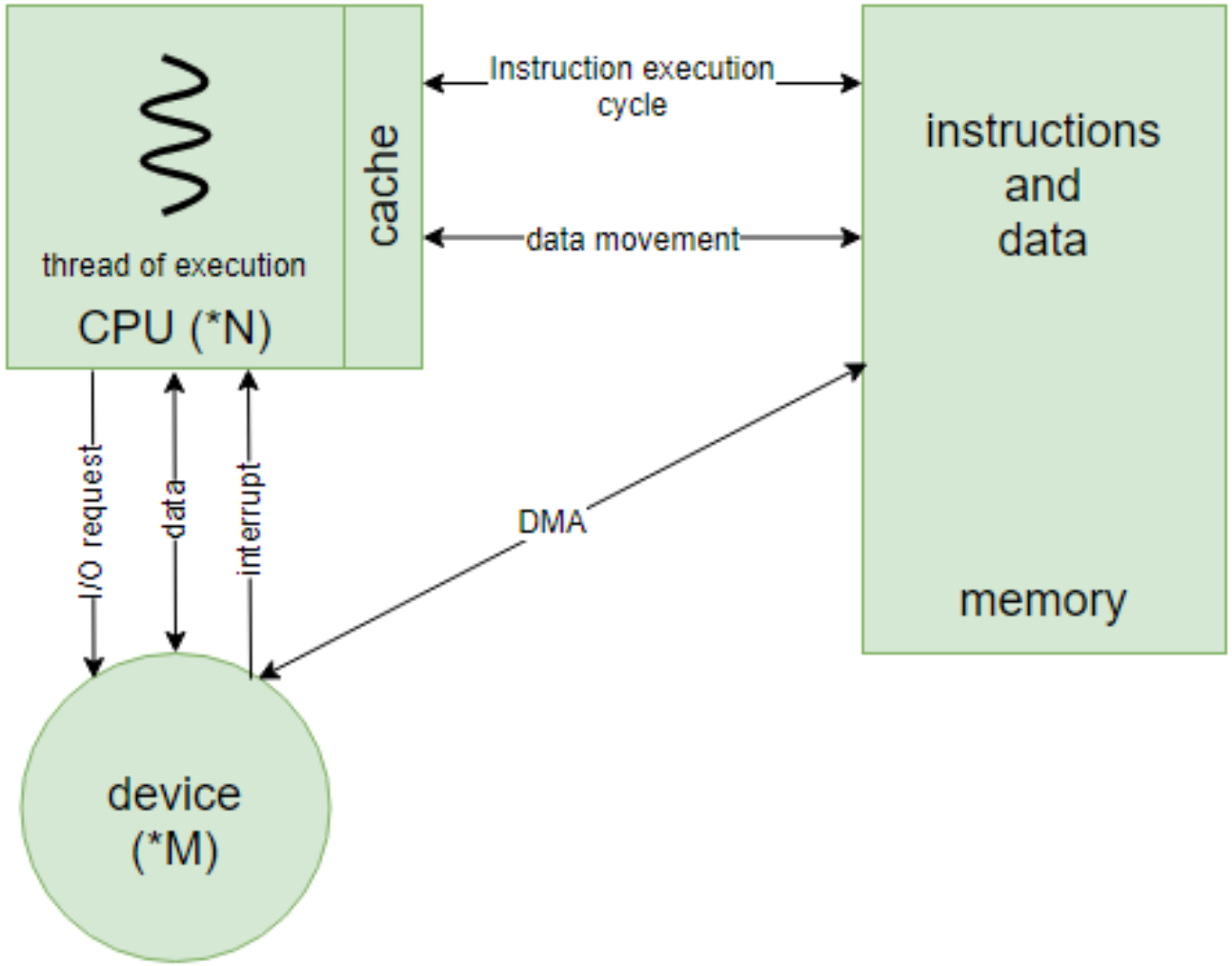
- **Important principle, performed at many levels in a computer (in hardware, operating system, software)**
- **Information in use copied from slower to faster storage temporarily**
- **Faster storage (cache) checked first to determine if information is there**
  - If it is, information used directly from the cache (fast)
  - If not, data copied to cache and used there
- **Cache smaller than storage being cached**
  - Cache management important design problem
  - Cache size and replacement policy

Computer-System Architecture

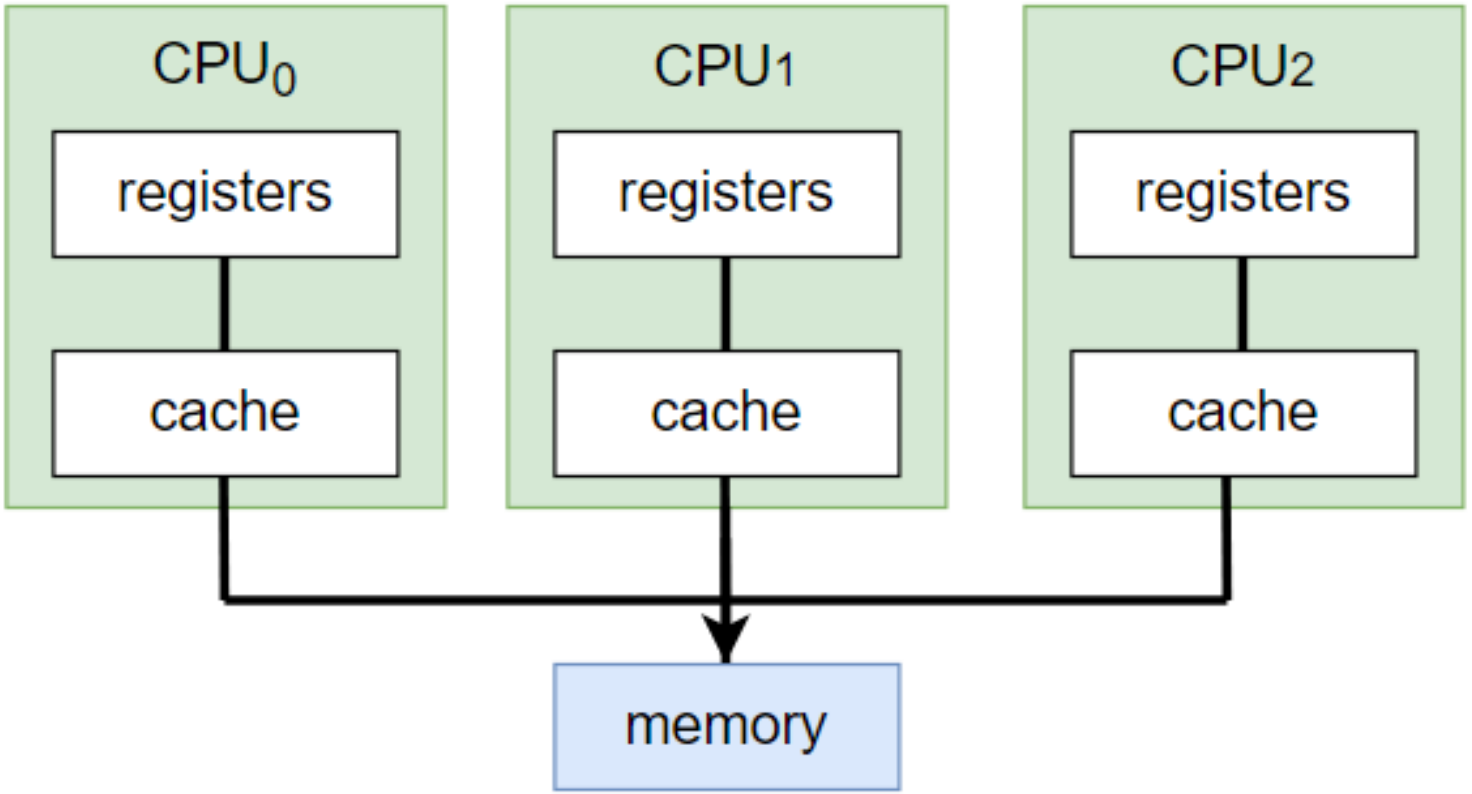
- **Most systems use a single general-purpose processor (PDAs through mainframes)**
  - Most systems have special-purpose processors as well

- **Multiprocessors systems growing in use and importance**
  - Also known as **parallel systems**, **tightly-coupled systems**
  - Advantages include:
    - **Increased throughput**
    - **Economy of scale**
    - **Increased reliability – graceful degradation** or **fault tolerance**
  - Two types:
    - **Asymmetric Multiprocessing**
    - **Symmetric Multiprocessing**

How a Modern Computer Works

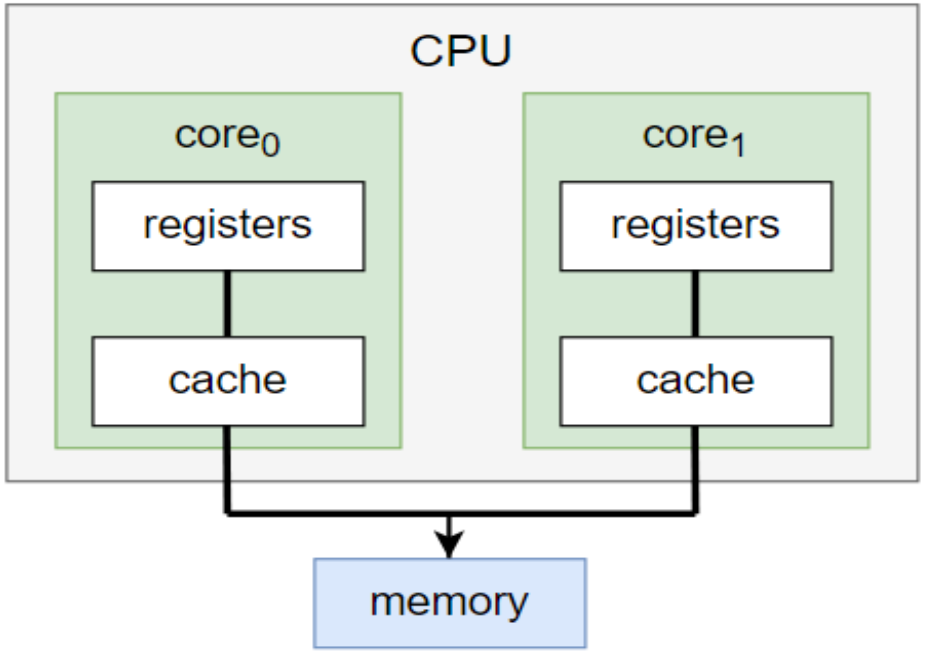


Symmetric Multiprocessing Architecture



A Dual-Core Design

- **UMA and NUMA architecture variations**
- **Multi-chip and multicore**
- **Systems containing all chips vs. blade servers**
  - Chassis containing multiple separate systems



Clustered Systems

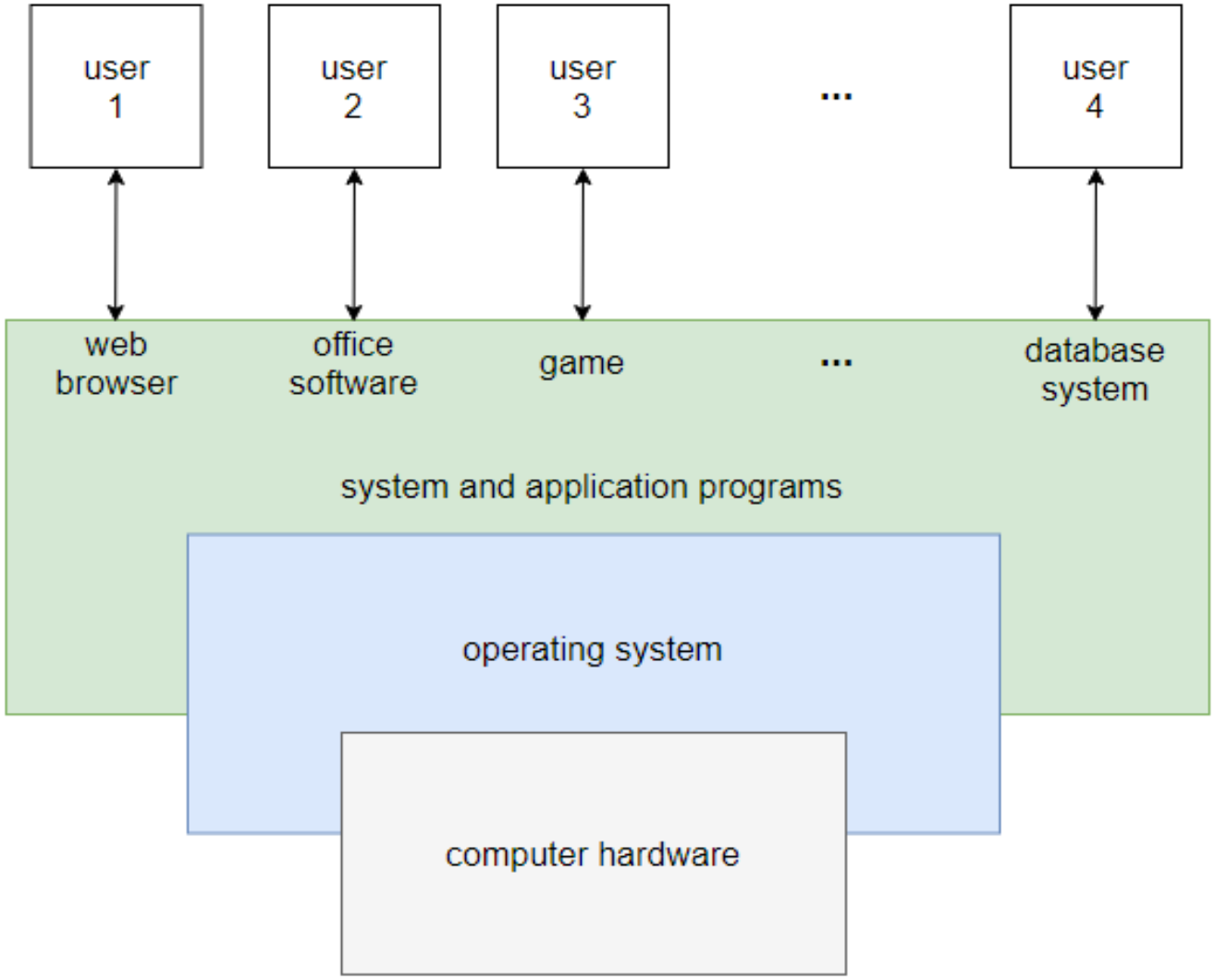
- **Like multiprocessor systems, but multiple systems working together**
  - Usually sharing storage via a **storage-area network (SAN)**
  - Provides a **high-availability** service which survives failures
    - **Asymmetric clustering** has one machine in hot-standby mode
    - **Symmetric clustering** has multiple nodes running applications, monitoring each other
  - Some clusters are for **high-performance computing (HPC)**
    - Applications must be written to use **parallelization**
  - Some have **distributed lock manager** (**DLM**) to avoid conflicting operations

What is an Operating System?

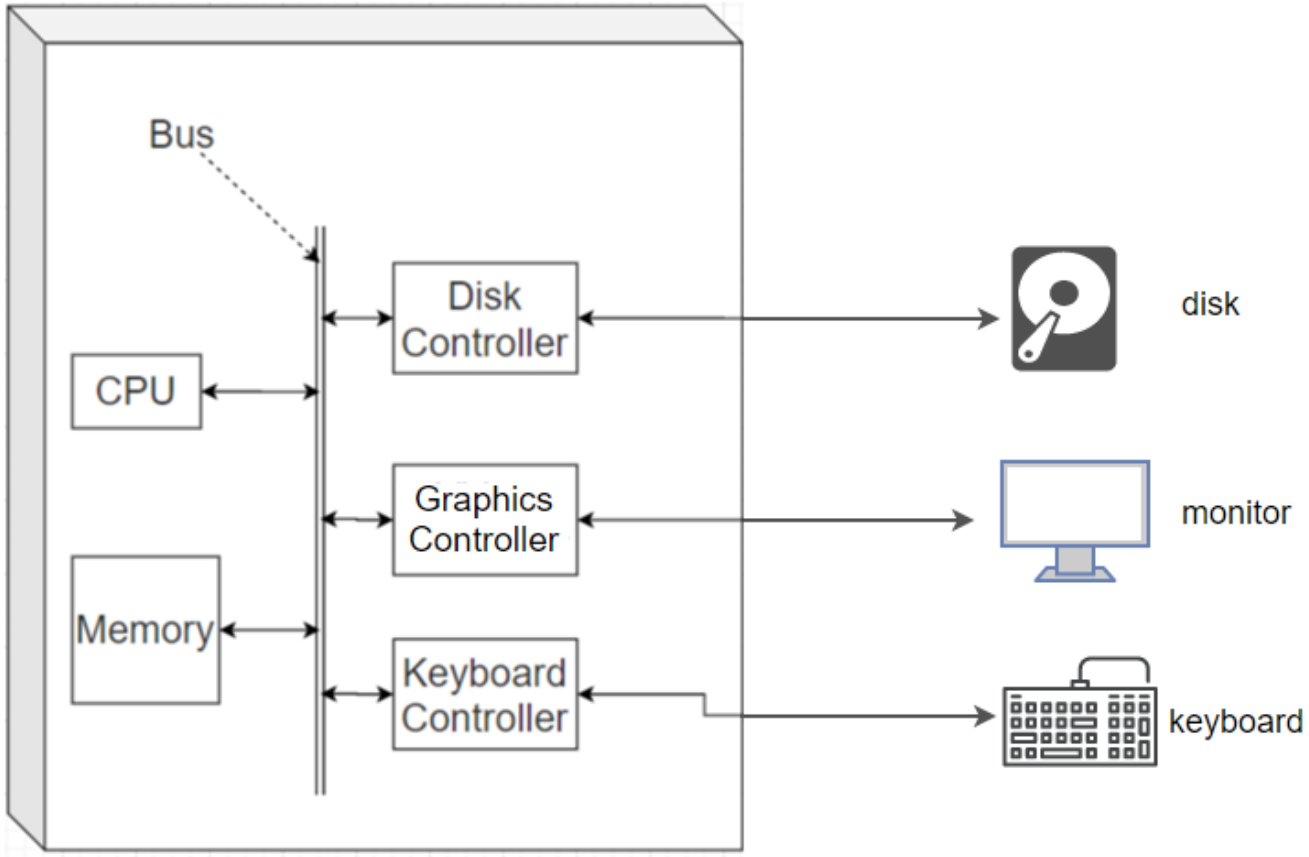
- **A program that acts as an intermediary between a user of a computer and the computer hardware**

- **Operating system goals:**
  - Execute user programs and make solving user problems easier
  - Make the computer system convenient to use
  - Use the computer hardware in an efficient manner

Four Components of a Computer System



Simplified PC Hardware



Why an OS

- **Games –OS looks like overkill**
- **Many common pieces**
  - reading button presses
- **Should all work alike**
- **Need advanced features**

Why an OS

- **Games need advanced features:**
  - Asynchronicity
  - Dynamic memory allocation
  - Timers
  - Game save/ restore
  - Switch games

Terms & Concepts

- **Device driver - a part of the OS that interfaces to a device controller.**
- **Kernel - part of an OS that implements basic functions – always running.**
- **Service - functions an OS provides to users through programmer interfaces.**

Who is the User?

- **Who is “we”?**
  - End users
  - Application programmers
  - System programmers
  - Administrators

End Users Want

- Ease of use and learn
- Adaptable to user's style
- Alternative ways to do things
- Lots of visual cues
- Free of nasty surprises
- Reliable quick response
- Consistency among features

Application Programmers Want

- **Easy to write programs**
  - read keystrokes
  - draw to screen
  - change windows
- **Consistent view of system**
- **Easy to use system facilities**
  - creating new windows
  - accessing a network
- **Platform portability**

System Programmers Want

- **Easy to**
  - create correct programs
  - debug
  - maintain
  - expand

Administrators Want

- Easy to add or remove devices
- Protect system and data
- Easy to upgrade
- Easy to manage user accounts
- Good & predictable average response
- Affordable

The System's View

- **Details of internal operation**
- **Information often secret**
- **Subsystem overviews for application programmers**

The System's View

- **This course is about how OSs:**
  - run a program
  - store a file
  - manage memory
  - etc.

Computer System Structure

- **Computer system can be divided into four components:**
  - Hardware – provides basic computing resources
    - CPU, memory, I/O devices
  - Operating system
    - Controls and coordinates use of hardware among various applications and users
  - Application programs – define the ways in which the system resources are used to solve the computing problems of the users
    - Word processors, compilers, web browsers, database systems, video games
  - Users
    - People, machines, other computers

What Operating Systems Do

- **Depends on the point of view**
- **Users want convenience, ease of use**
  - Don't care about **resource utilization**
- **But shared computer such as mainframe or minicomputer must keep all users happy**
- **Users of dedicate systems such as workstations have dedicated resources but frequently use shared resources from servers**
- **Handheld computers are resource poor, optimized for usability and battery life**
- **Some computers have little or no user interface, such as embedded computers in devices and automobiles**

Operating System Definition

- **OS is a resource allocator**
  - Manages all resources
  - Decides between conflicting requests for efficient and fair resource use

- **OS is a control program**
  - Controls execution of programs to prevent errors and improper use of the computer

Operating System Definition (Cont.)

- **No universally accepted definition**
- **"Everything a vendor ships when you order an operating system" is good approximation**
  - But varies wildly
- **"The one program running at all times on the computer" is the kernel. Everything else is either a system program (ships with the operating system) or an application program.**

Computer Startup

- **bootstrap program is loaded at power-up or reboot**
  - Typically stored in ROM or EPROM, generally known as **firmware**
  - Initializes all aspects of system
  - Loads operating system kernel and starts execution

Resources Managed by the OS

- CPU
- Main memory & cache
- Secondary storage
- I/O devices
- File system
- User interfaces
- Network access
- Protection & security

Major Modules of an OS

- **Process management**
- **CPU scheduling**
- **Memory & cache management**
- **File management**
- **I/O management**
- **Disk scheduling**
- **GUI management**
- **Network management**
- **Protection & security**

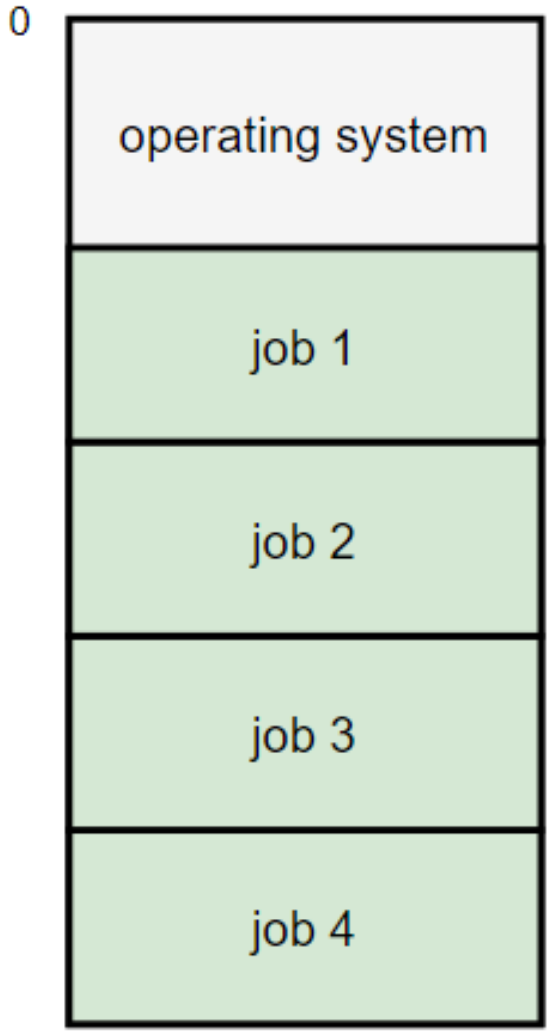
Operating System Structure(1)

- **Multiprogramming needed for efficiency**
  - Single user cannot keep CPU and I/O devices busy at all times
  - Multiprogramming organizes jobs (code and data) so CPU always has one to execute
  - A subset of total jobs in system is kept in memory
  - One job selected and run via **job scheduling**
  - When it has to wait (for I/O for example), OS switches to another job

Operating System Structure(2)

- **Timesharing (multitasking) is logical extension in which CPU switches jobs so frequently that users can interact with each job while it is running, creating interactive computing**
  - **Response time** should be < 1 second
  - Each user has at least one program executing in memory ⇨**process**
  - If several jobs ready to run at the same time ⇨ **CPU scheduling**
  - If processes don't fit in memory, **swapping** moves them in and out to run
  - **Virtual memory** allows execution of processes not completely in memory

Memory Layout for Multiprogramming System



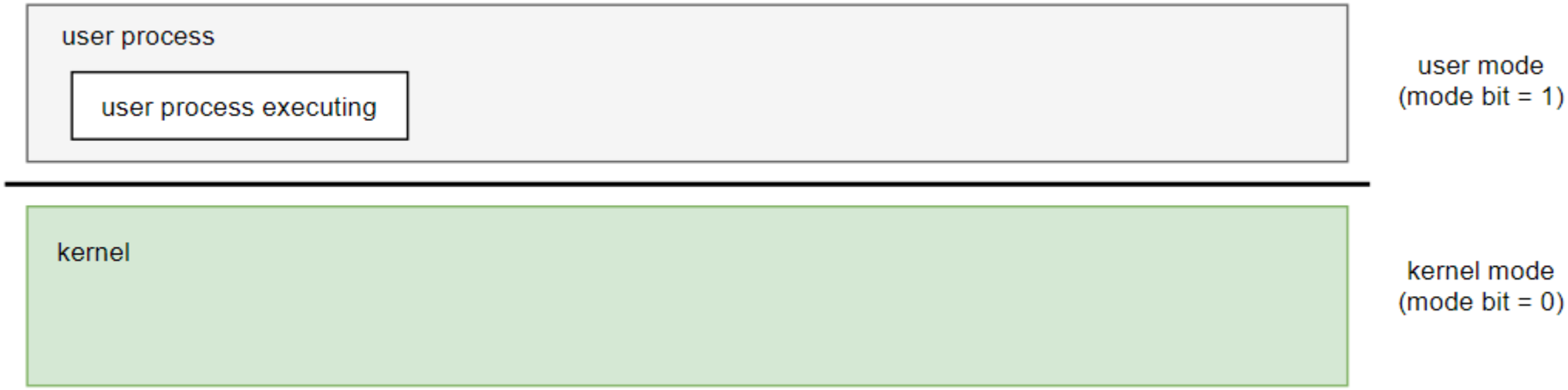
Operating-System Operations(1)

- **Interrupt driven by hardware**
- **Software error or request creates exception or trap**
  - Division by zero, request for operating system service
- **Other process problems include infinite loop, processes modifying each other or the operating system**

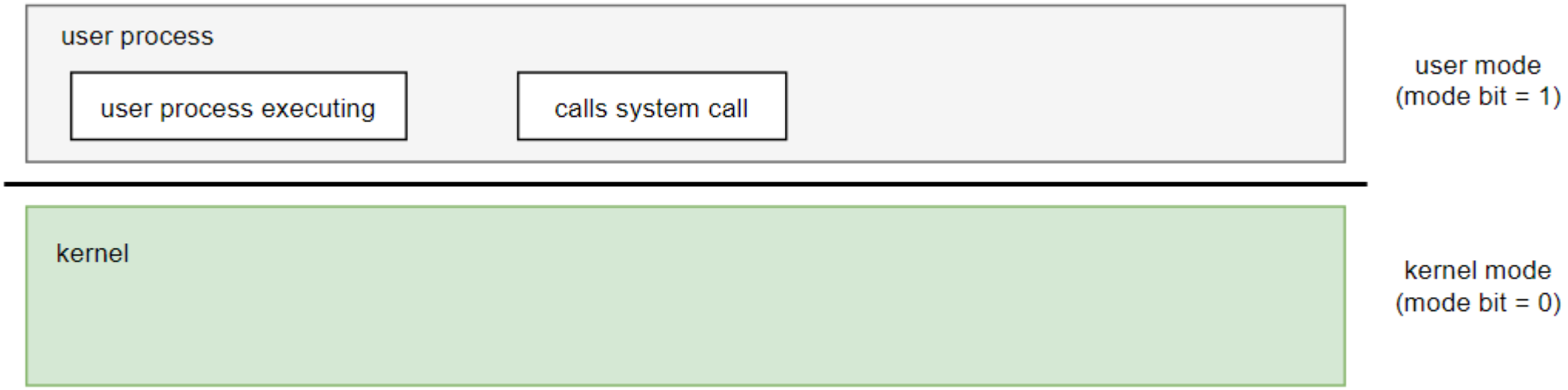
Operating-System Operations(2)

- **Dual-mode operation allows OS to protect itself and other system components**
  - **User mode** and **kernel mode**
  - **Mode bit** provided by hardware
    - Provides ability to distinguish when system is running user code or kernel code
    - Some instructions designated as **privileged**, only executable in kernel mode
    - System call changes mode to kernel, return from call resets it to user

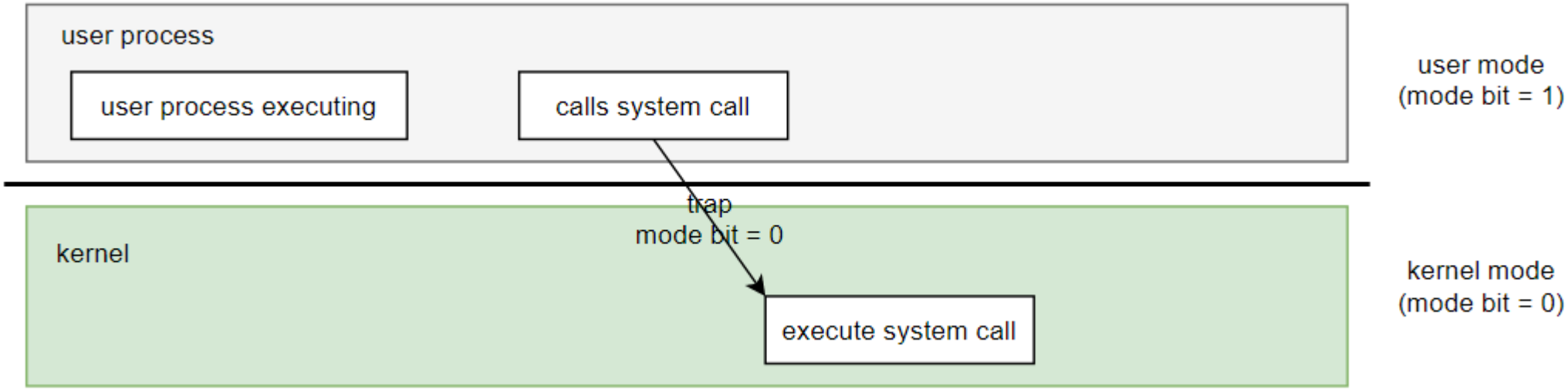
Transition from User to Kernel Mode



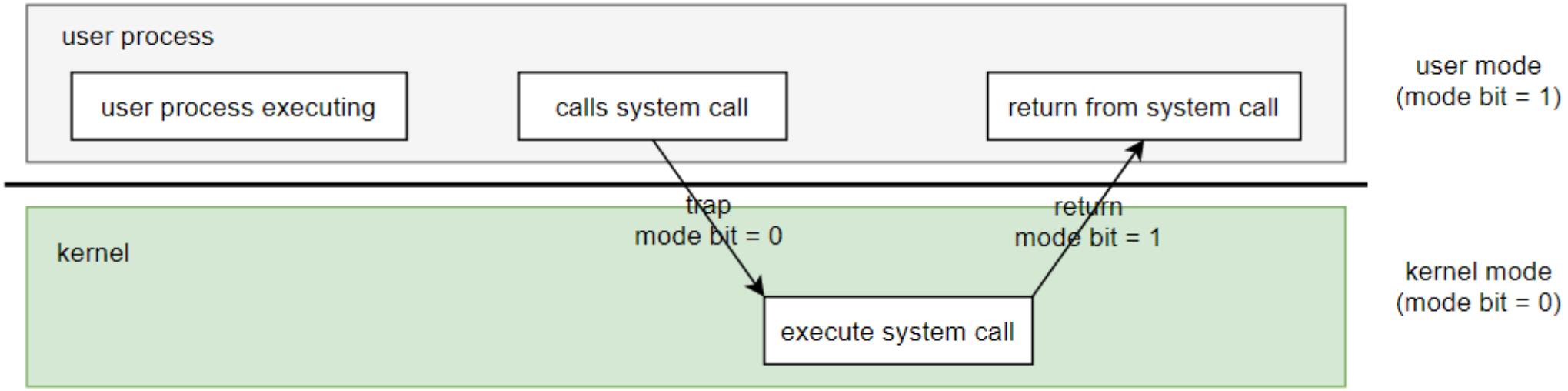
Transition from User to Kernel Mode



Transition from User to Kernel Mode



Transition from User to Kernel Mode



Process Management

- **A process is a program in execution. It is a unit of work within the system. Program is a *passive entity*, process is an *active entity*.**
- **Process needs resources to accomplish its task**
  - CPU, memory, I/O, files
  - Initialization data
- **Process termination requires reclaim of any reusable resources**
- **Typically system has many processes, some user, some operating system running concurrently on one or more CPUs**
  - Concurrency by multiplexing the CPUs among the processes

Process Management Activities

The operating system is responsible for the following activities in connection with process management:

- **Creating and deleting both user and system processes**
- **Suspending and resuming processes**
- **Providing mechanisms for process synchronization**
- **Providing mechanisms for process communication**
- **Providing mechanisms for deadlock handling**

Memory Management

- **All data in memory before and after processing**
- **All instructions in memory in order to execute**
- **Memory management determines what is in memory when**
  - Optimizing CPU utilization and computer response to users
- **Memory management activities**
  - Keeping track of which parts of memory are currently being used and by whom
  - Deciding which processes (or parts thereof) and data to move into and out of memory
  - Allocating and deallocating memory space as needed

Storage Management

- **OS provides uniform, logical view of information storage**
  - Abstracts physical properties to logical storage unit - **file**
- **File-System management**
  - Files usually organized into directories
  - Access control on most systems to determine who can access what
  - OS activities include
    - Creating and deleting files and directories
    - Primitives to manipulate files and directories
    - Mapping files onto secondary storage
    - Backup files onto stable (non-volatile) storage media

Mass-Storage Management(1)

- **Usually disks used to store data that does not fit in main memory or data that must be kept for a “long” period of time**
- **Proper management is of central importance**
- **Entire speed of computer operation hinges on disk subsystem and its algorithms**

Mass-Storage Management(2)

- **OS activities**
  - Free-space management
  - Storage allocation
  - Disk scheduling
- **Some storage need not be fast**
  - Tertiary storage includes optical storage, magnetic tape
  - Still must be managed – by OS or applications
  - Varies between WORM (write-once, read-many-times) and RW (read-write)

Performance of Various Levels of Storage

Level	1	2	3	4	5
Name	register	cache	main memory	solid state disk	magnetic disk
Typical size	<1KB	<16MB	<64GB	<8TB	<20TB
Implementation technology	custom memory with multiple ports CMOS	on-chip or off-chip CMOS SRAM	CMOS SRAM	flash memory	magnetic disk
Access time(ns)	0.25 - 0.5	0.5 - 20	30 - 200	25,000 - 50,000	5,000,000
Bandwidth(GB/sec)	1,000-10,000	200 – 1,000	10 – 100	0.5 - 2	0.03 – 0.3
Managed by	compiler	hardware	operating system	operating system	operating system
Backed by	cache	main memory	disk	disk	disk or tape

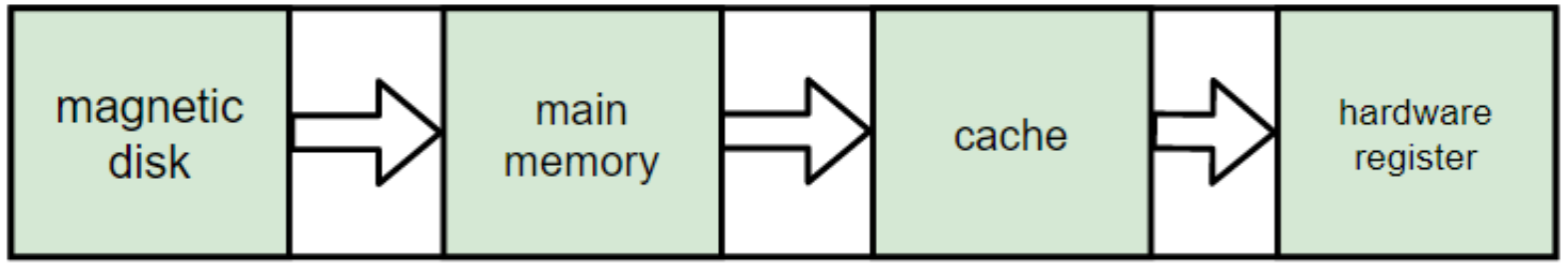
(2020 updated)

- **Movement between levels of storage hierarchy can be explicit or implicit**

https://www.memorybenchmark.net/
https://www.harddrivebenchmark.net/

Migration of Integer A from Disk to Register

- Multitasking environments must be careful to use most recent value, no matter where it is stored in the storage hierarchy



- Multiprocessor environment must provide cache coherency in hardware such that all CPUs have the most recent value in their cache
- Distributed environment situation even more complex

I/O Subsystem

- **One purpose of OS is to hide peculiarities of hardware devices from the user**
- **I/O subsystem responsible for**
  - Memory management of I/O including buffering (storing data temporarily while it is being transferred), caching (storing parts of data in faster storage for performance), spooling (the overlapping of output of one job with input of other jobs)
  - General device-driver interface
  - Drivers for specific hardware devices

Protection & Security

- **Protection – any mechanism for controlling access of processes or users to resources defined by the OS**

- **Security – defense of the system against internal and external attacks**
  - Huge range, including denial-of-service, worms, viruses, identity theft, theft of service

Types of OSs

- **Single-user**
- **Multi-tasking or multi-programming**
- **Time sharing**
- **Network and distributed**
- **Real-time**

Single-user OS

- One user
- One process running
- Limited memory management
- File services
- I/O services

Multi-tasking OS

- **Multiple processes running**
- **May still be single user**
- **All that a single task system does, +**
- **CPU scheduling**
  - Context switching

Multi-user OS

- **Multiple users**
- **Users are interactive**
- **Jobs**
  - are short
  - need fewer resources
  - need fast response time

Network/ Distributed OS

- **Low-level services**
  - Connect to the network
  - Send messages between systems
- **Higher-level services**
  - Browsing
  - File sharing
  - Print services

Real-Time OS

- **Time deadlines**
- **Hard deadlines**
  - Useless results or catastrophe
- **Soft deadlines**
  - Some lateness tolerated

What should an OS do?

- **ALL I/O routines?**
  - Even for devices not on this machine?
- **Mathematics routines?**
  - Even if not used in this application?
- **GUI interface?**
  - In a shell instead?
- **As much as possible?**
  - As little as possible?

Computing Environments - Traditional

- Stand-alone general purpose machines
- But blurred as most systems interconnect with others (i.e. the Internet)
- Portals provide web access to internal systems
- Network computers (thin clients) are like Web terminals
- Mobile computers interconnect via wireless networks
- Networking becoming ubiquitous – even home systems use firewalls to protect home computers from Internet attacks

Computing Environments - Mobile

- **Handheld smartphones, tablets, etc**
- **What is the functional difference between them and a "traditional" laptop?**
- **Extra feature – more OS features (GPS, gyroscope)**
- **Allows new types of apps like *augmented reality***
- **Use WiFi, or cellular data networks for connectivity**
- **Leaders are Apple iOS and Google Android**

Computing Environments – Distributed

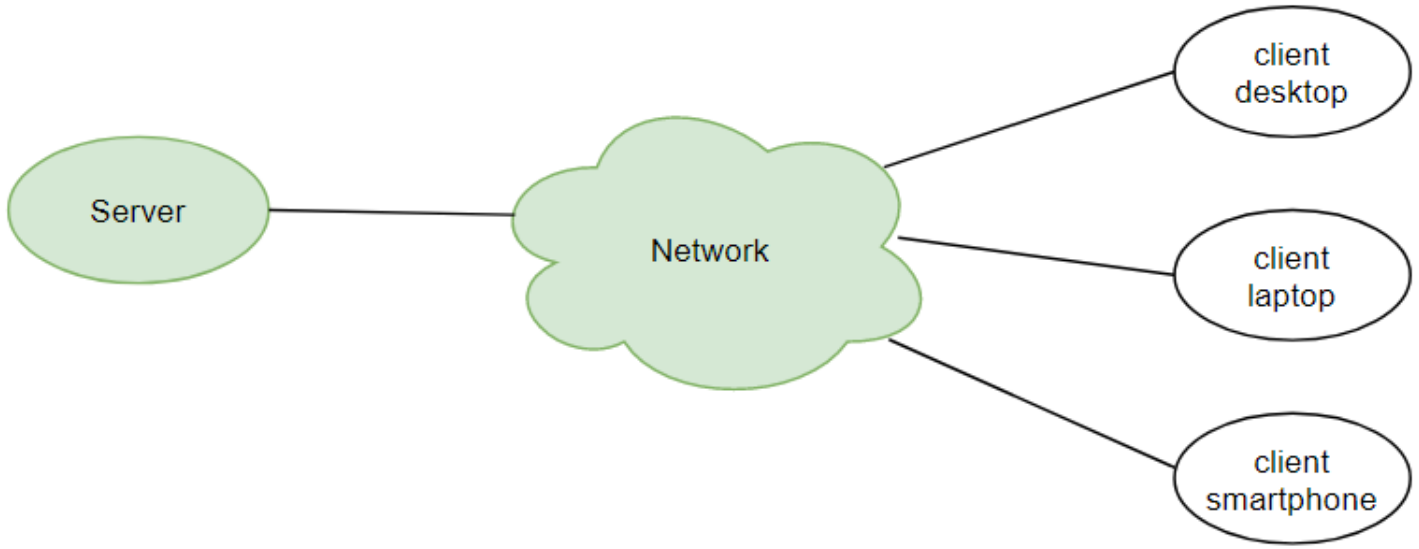
Distributed

- Collection of separate, possibly heterogeneous, systems networked together
  - **Network** is a communications path, **TCP/IP** most common
    - **Local Area Network** (**LAN**)
    - **Wide Area Network** (**WAN**)
    - **Metropolitan Area Network** (**MAN**)
    - **Personal Area Network** (**PAN**)
- **Network Operating System** provides features between systems across network
  - Communication scheme allows systems to exchange messages
  - Illusion of a single system

Computing Environments – Client-Server

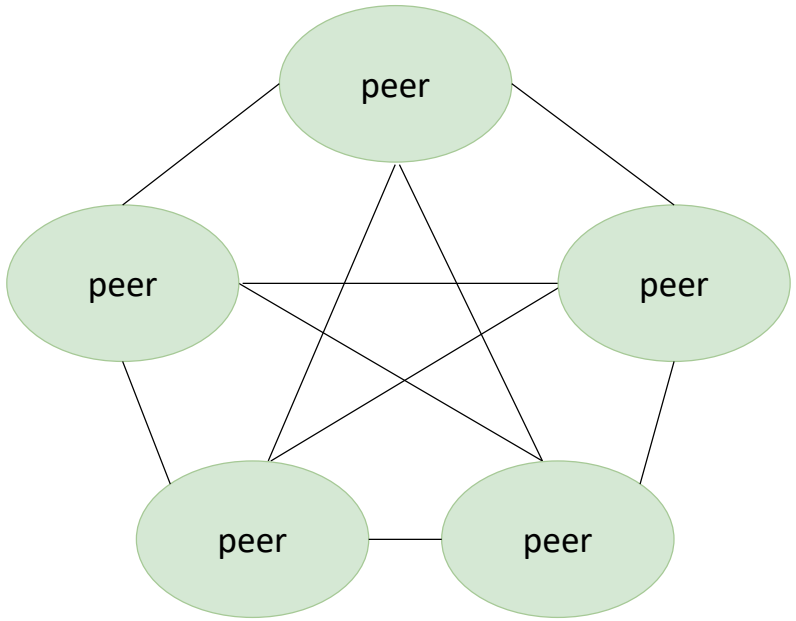
Client-Server Computing

- Dumb terminals supplanted by smart PCs
- Many systems now **servers**, responding to requests generated by **clients**
  - **Compute-server system** provides an interface to client to request services (i.e., database)
  - **File-server system** provides interface for clients to store and retrieve files



Computing Environments - Peer-to-Peer

- **Another model of distributed system**
- **P2P does not distinguish clients and servers**
  - Instead all nodes are considered peers
  - May each act as client, server or both
  - Examples include BitComet, BitTorrent, **Voice over IP** *(***VoIP***)* such as Skype



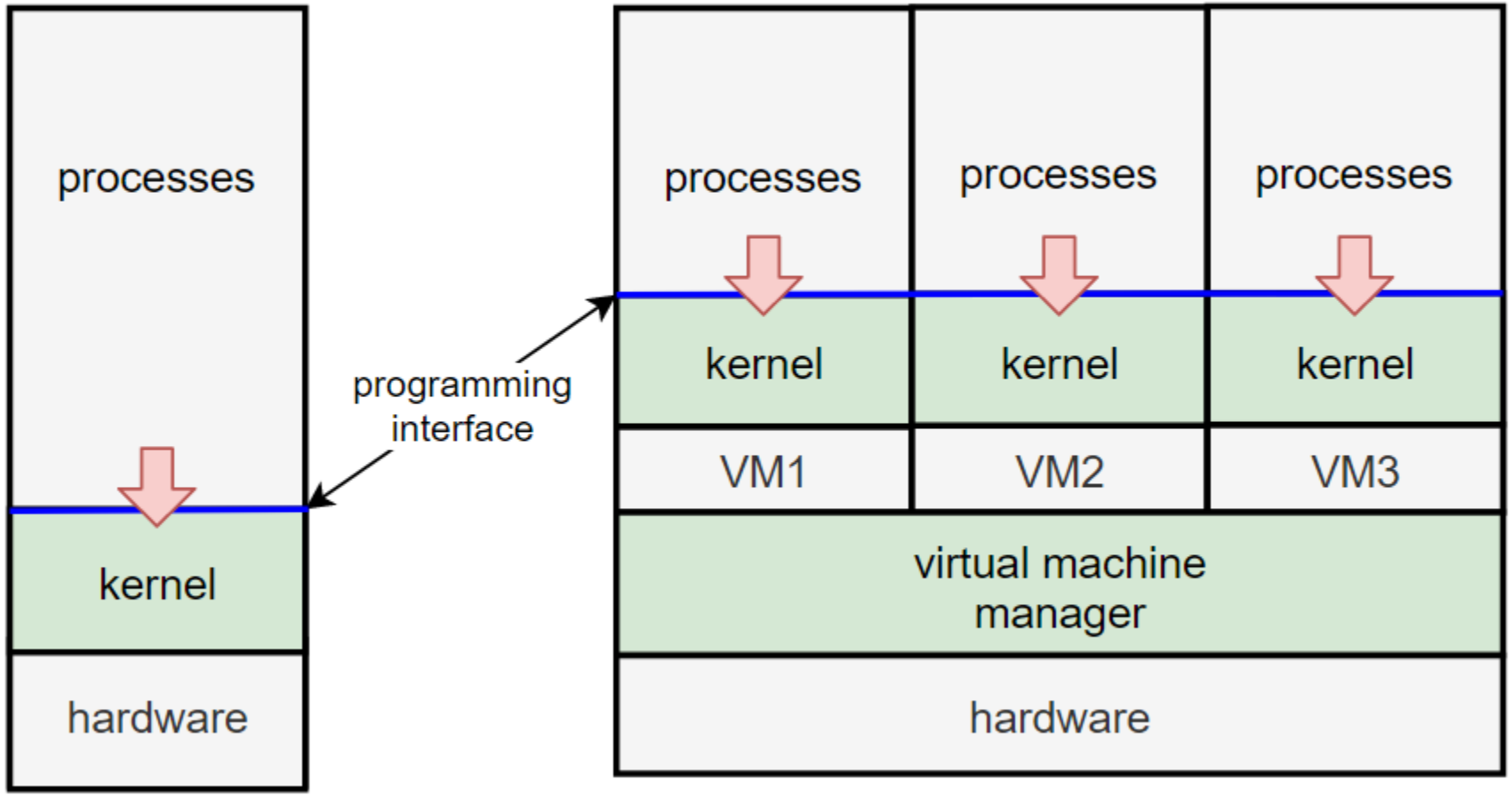
Computing Environments - Virtualization

- **Allows operating systems to run applications within other OSes**
  - Vast and growing industry
- **Emulation used when source CPU type different from target type (i.e. Intel x86 to ARM)**
  - Generally slowest method
  - When computer language not compiled to native code – **Interpretation**
- **Virtualization – OS natively compiled for CPU, running guest Oses also natively compiled**
  - Consider Vmware running Windows guests, each running applications, all on native Windows **host** OS
  - **VM** provides virtualization services

Computing Environments - Virtualization

- **Use cases involve laptops and desktops running multiple OSes for exploration or compatibility**
  - Apple laptop running macOS host, Windows as a guest
  - Developing apps for multiple OSes without having multiple systems
  - QA testing applications without having multiple systems
  - Executing and managing compute environments within data centers
- **VM can run natively, in which case they are also the host**

Computing Environments - Virtualization



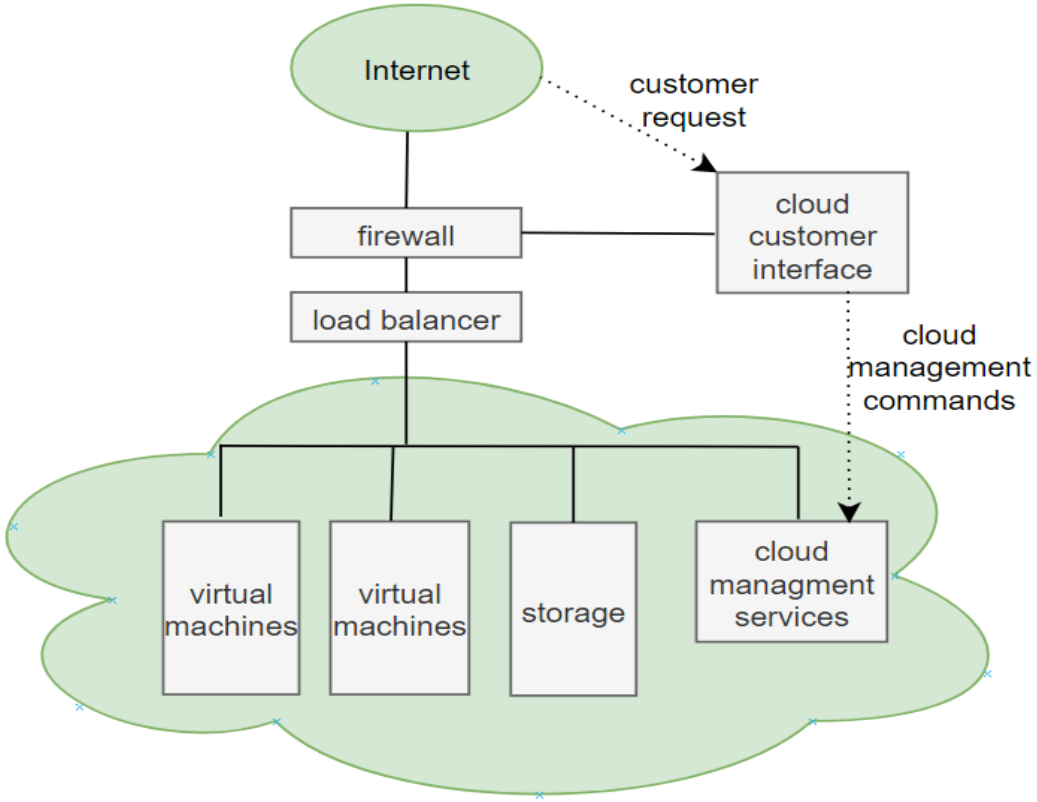
Computing Environments – Cloud Computing

- **Delivers computing, storage, even apps as a service across a network**
- **Logical extension of virtualization as based on virtualization**
  - Amazon **EC2** has thousands of servers, millions of VMs, PBs of storage available across the Internet, pay based on usage
- **Many types**
  - **Public cloud** – available via Internet to anyone willing to pay
  - **Private cloud** – run by a company for the company's own use
  - **Hybrid cloud** – includes both public and private cloud components

Computing Environments – Cloud Computing

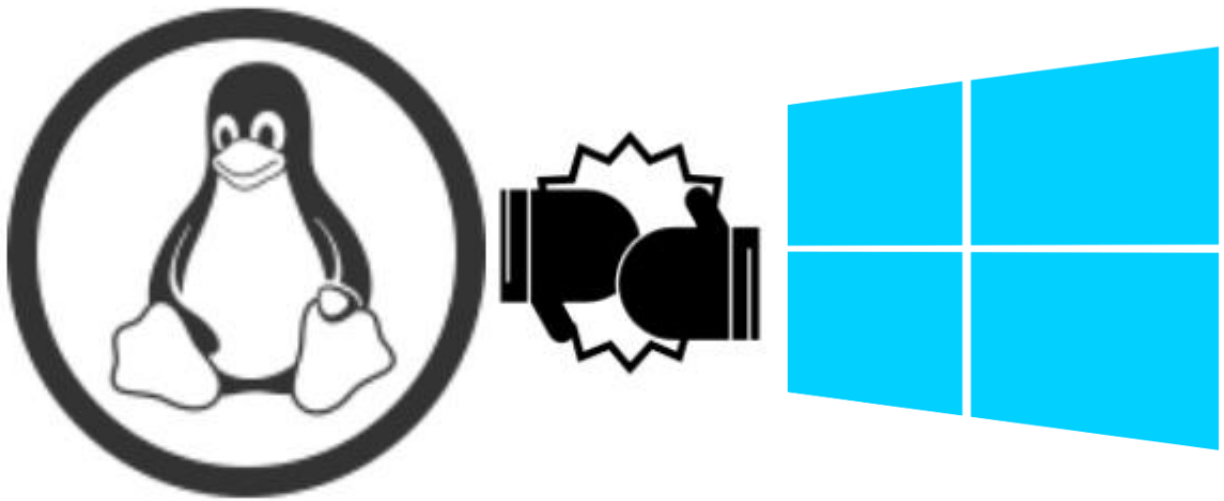
Cloud compute environments composed of traditional OSes, plus VMs, plus cloud management tools

- Internet connectivity requires security like firewalls
- Load balancers spread traffic across multiple applications



Open-Source Operating Systems

- **Operating systems made available in source-code format rather than just binary closed-source**
- **Counter to the copy protection and Digital Rights Management (DRM) movement**
- **Started by Free Software Foundation (FSF), which has "copyleft" GNU Public License (GPL)**
- **Examples include GNU/Linux and BSD UNIX (including core of macOS), and many more**
- **Can use VM like VMware Player (Free on Windows), Virtualbox (open source and free on many platforms - https://www.virtualbox.org/)**
  - Use to run guest operating systems for exploration



Linux vs. Windows … 10 diffs

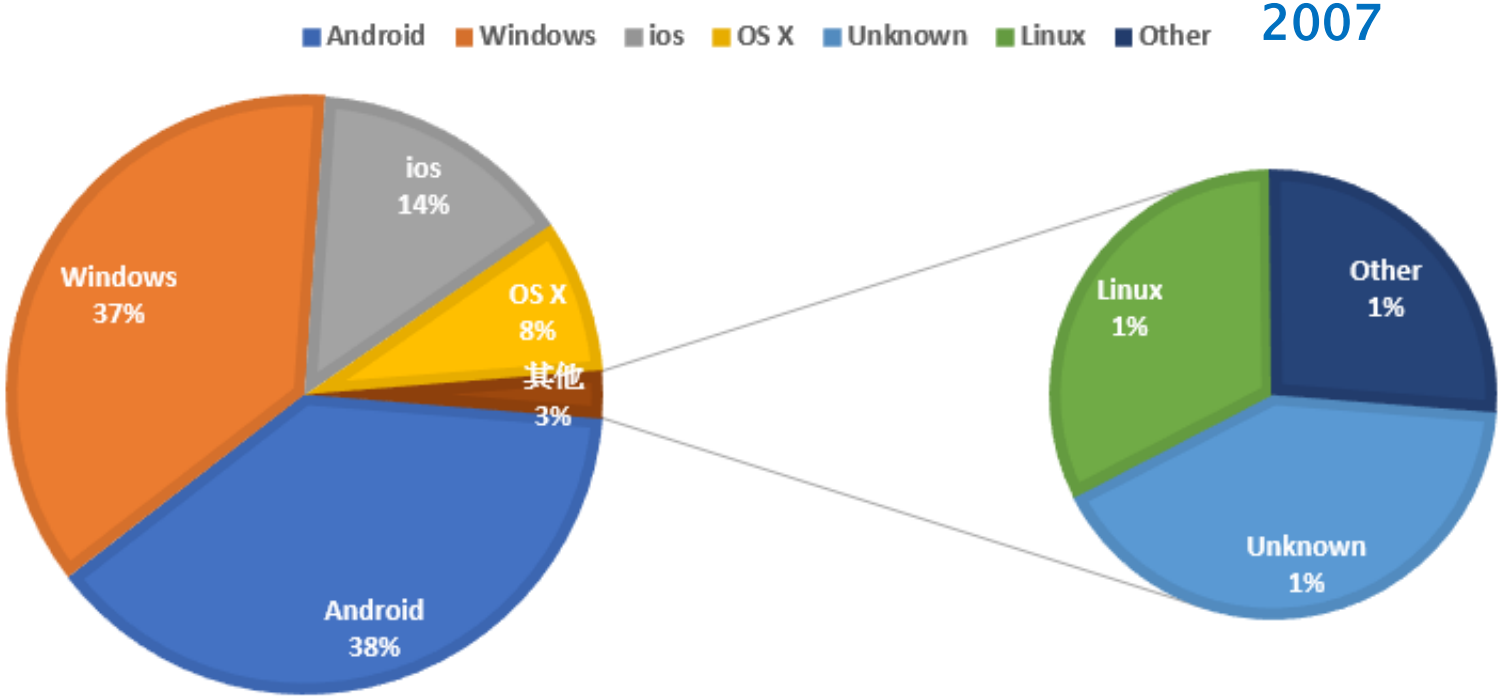
1. Open source operating system
2. Various distributions
3. Proven Security
4. More efficient
5. More reliable
6. Customizable
7. Transparent vs Propriety
8. Support via a huge community of user forums
9. Formats are free, freedom is preserved
10. Use the extra cash for rewards





OS Market Share

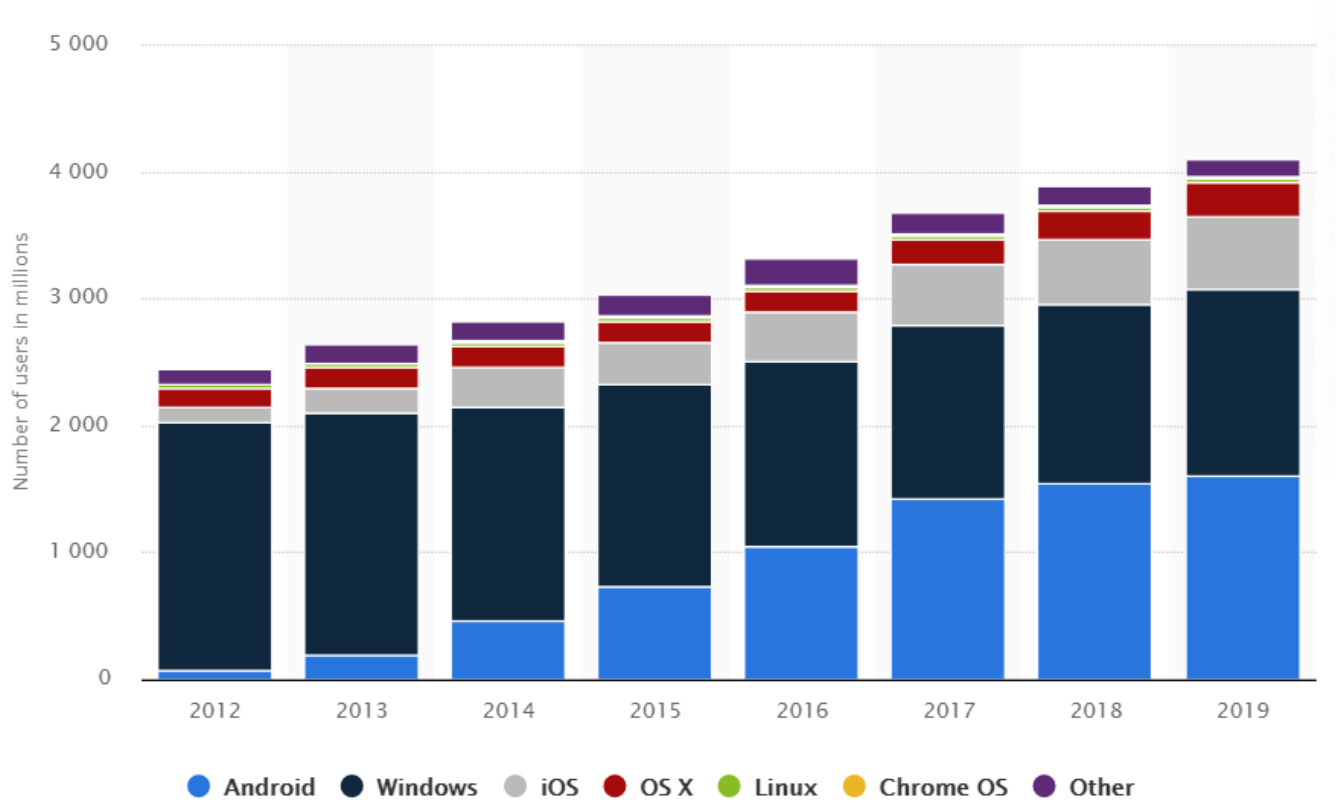
As of November 2007



As of June 2020

Source：https://gs.statcounter.com/os-market-share

Population of internet users worldwide by operating system *(in millions)*



from 2012 to 2019

Source : https://www.statista.com/statistics/543185/worldwide-internet-connected-operating-system-population/



What is the future of OS?

Cloud Computing !!!

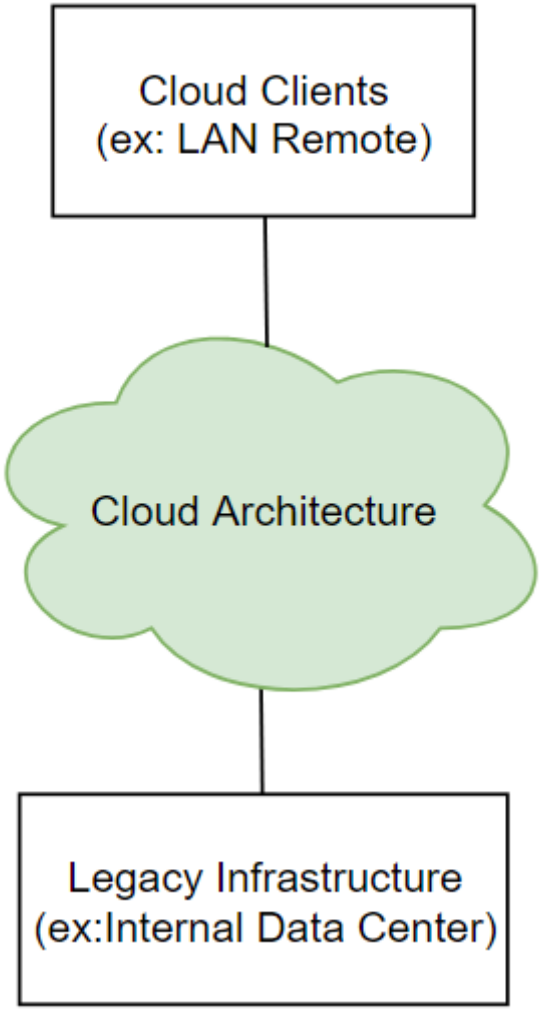


Cloud Computing

A paradigm in which information is permanently stored in servers on the Internet and cached temporarily on clients that include desktops, entertainment centers, table computers, notebooks, wall computers, handhelds, etc.

--IEEE Computer Society

Cloud Architecture



PaaS: Platform as a Service

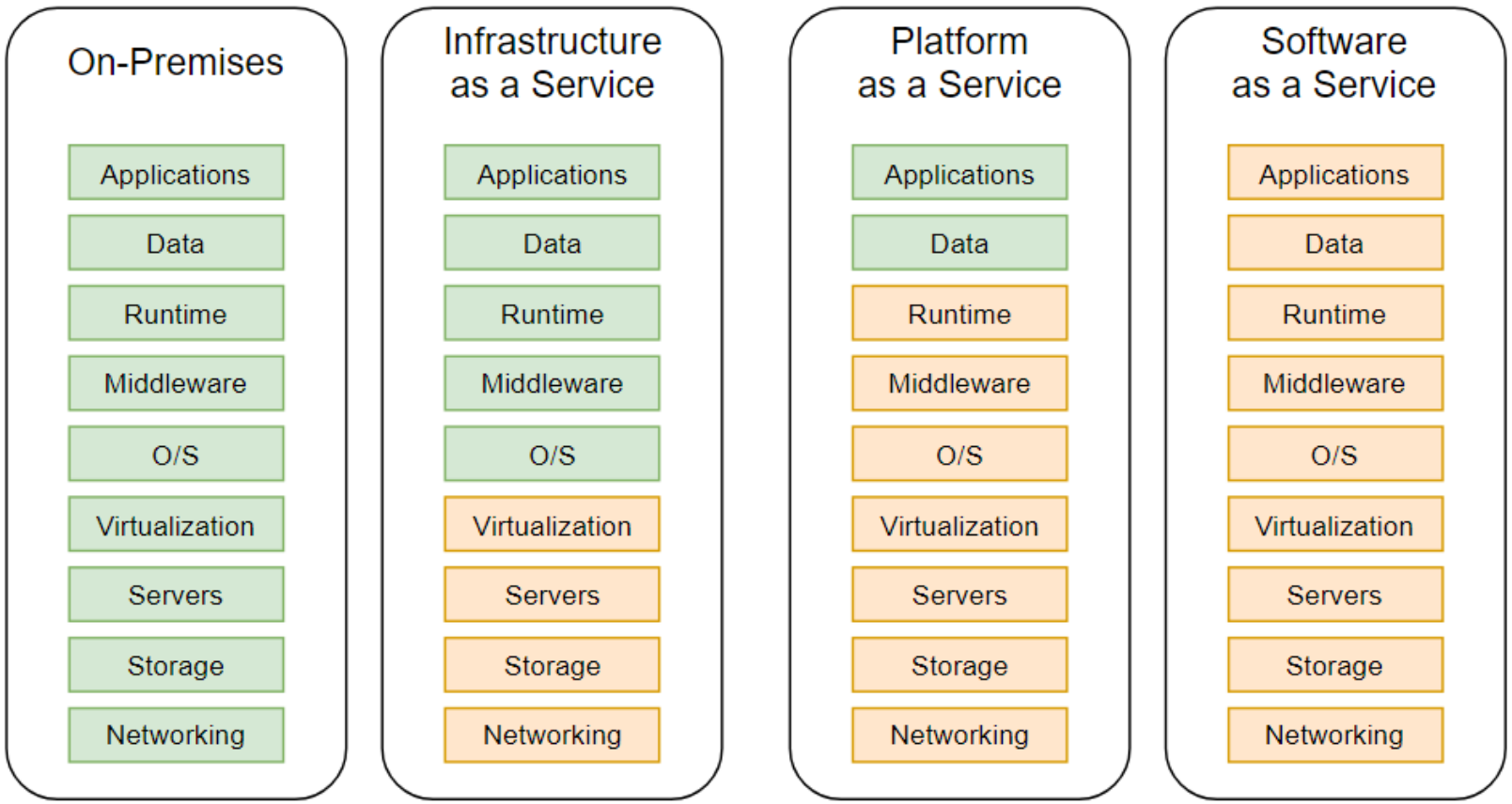
- **Provides a platform for software creation.**
  - Allows businesses to design and create applications that are built into the PaaS with special software components.
- **Giving developers the freedom to concentrate on building the software without having to worry about operating systems, software updates, storage, or infrastructure.**
  - All servers, storage, and networking can be managed by the enterprise or a third-party provider while the developers can maintain management of the applications.

SaaS: Software as a Service

- **SaaS utilizes the internet to deliver applications to its users.**
  - A majority of SaaS applications run directly through your web browser, which means they do not require any downloads or installations on the client side.
- **SaaS eliminates the need to have IT staff download and install applications on each individual computer.**
  - With SaaS, vendors manage all potential technical issues, such as data, middleware, servers, and storage, resulting in streamlined maintenance and support for the business.
- **SaaS provides numerous advantages to employees and companies by greatly reducing the time and money spent on tedious tasks such as installing, managing, and upgrading software.**

IaaS: Infrastructure as a Service

- **Allows businesses to purchase resources on-demand and as-needed instead of having to buy hardware outright.**
  - IaaS delivers cloud computing infrastructure, including servers, network, operating systems, and storage, through virtualization technology.
- **Provides the same technologies and capabilities as a traditional data center without having to physically maintain or manage all of it.**
  - IaaS clients can still access their servers and storage directly, but it is all outsourced through a “virtual data center” in the cloud.


On-Premises
Applications
Data
Runtime
Middleware
O/S
Virtualization
Servers
Storage
Networking
Infrastructure as a Service
Applications
Data
Runtime
Middleware
O/S
Virtualization
Servers
Storage
Networking
Platform as a Service
Applications
Data
Runtime
Middleware
O/S
Virtualization
Servers
Storage
Networking
Software as a Service
Applications
Data
Runtime
Middleware
O/S
Virtualization
Servers
Storage
Networking
Your Manage
Other Manages

Common Examples of SaaS, PaaS, & IaaS

Platform Type	Common Examples
SaaS	BigCommerce, Google Apps, Salesforce, Dropbox, MailChimp, ZenDesk, DocuSign, Slack, Hubspot.
PaaS	AWS Elastic Beanstalk, Heroku, Windows Azure (mostly used as PaaS), Force.com, OpenShift, Apache Stratos, Magento Commerce Cloud.
IaaS	Amazon Web Services (AWS), Microsoft Azure, DigitalOcean, Linode, Rackspace, Cisco Metapod, Google Compute Engine (GCE).

(2019)

Source：https://blog.technavio.com/blog/saas-paas-iaas



Thanks for listening